

.NET Full Stack Developer Interview Preparation

200 Logical Programming Questions (Topic Wise)

Experience Level: 5–10 Years .NET Developers

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1. Number Programs (25 Questions)

Easy

1. Check Even or Odd Number
2. Check Positive, Negative, or Zero
3. Find Largest of Two Numbers
4. Find Largest of Three Numbers
5. Swap Two Numbers
6. Reverse a Number
7. Count Digits in a Number
8. Sum of Digits
9. Product of Digits
10. Find Power of a Number

Medium

1. Check Palindrome Number
2. Check Prime Number
3. Print Prime Numbers in a Range
4. Find Factorial
5. Find GCD (HCF)

6. Find LCM
7. Generate Fibonacci Series
8. Find Nth Fibonacci Number
9. Convert Decimal to Binary
10. Convert Binary to Decimal

Hard

1. Check Armstrong Number
 2. Check Perfect Number
 3. Check Strong Number
 4. Check Automorphic Number
 5. Check Leap Year
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2. Pattern Programs (25 Questions)

Star Patterns

1. Right Triangle Pattern
2. Inverted Triangle Pattern
3. Pyramid Pattern
4. Reverse Pyramid Pattern
5. Diamond Pattern
6. Hollow Diamond Pattern
7. Hollow Square Pattern
8. Solid Square Pattern

Number Patterns

1. Floyd's Triangle
2. Pascal Triangle
3. Number Pyramid
4. Numeric Triangle
5. Binary Pattern

Character Patterns

1. Alphabet Pyramid
2. Character Triangle

Advanced Patterns

1. Butterfly Pattern
2. Sandglass Pattern

3. Zig-Zag Pattern
 4. Hourglass Pattern
 5. X Pattern
 6. Cross Pattern
 7. Hollow Pyramid
 8. Hollow Rectangle
 9. Spiral Pattern
 10. Snake Pattern
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3. String Programs (30 Questions)

Basic

1. Reverse String
2. Check Palindrome String
3. Count Vowels
4. Count Consonants
5. Count Words
6. Count Characters
7. Remove Spaces
8. Convert Uppercase to Lowercase
9. Capitalize First Letter of Each Word
10. Replace Character

Intermediate

1. Count Occurrences of a Character
2. Find Substring
3. Check Substring Exists
4. Sort Characters in String
5. Find Longest Word
6. Find Shortest Word
7. Reverse Words in Sentence
8. Remove Special Characters
9. Remove Duplicate Characters
10. Find Duplicate Characters

Advanced

1. Check Anagram
2. Find First Non-Repeating Character
3. Find First Repeating Character
4. Count Frequency of Each Character
5. String Compression
6. Check Rotation of String

7. Check Valid Parentheses
 8. Find Common Characters in Two Strings
 9. Check Isomorphic Strings
 10. Longest Palindrome Substring
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4. Array Programs (35 Questions)

Basic

1. Find Largest Element
2. Find Smallest Element
3. Find Second Largest Element
4. Find Second Smallest Element
5. Reverse Array
6. Sort Array
7. Linear Search
8. Binary Search
9. Check Arrays Equal
10. Find Frequency of Elements

Intermediate

1. Remove Duplicates
2. Find Duplicate Elements
3. Find Missing Number
4. Find Common Elements
5. Merge Two Arrays
6. Rotate Array Left
7. Rotate Array Right
8. Separate Even and Odd Numbers
9. Move Zeros to End
10. Find Pair with Given Sum

Advanced

1. Find Triplet with Given Sum
2. Find Majority Element
3. Find Equilibrium Index
4. Find Leaders in Array
5. Find Union of Arrays
6. Find Intersection of Arrays
7. Kadane's Algorithm
8. Find Kth Largest Element
9. Find Kth Smallest Element
10. Stock Buy and Sell

Expert

1. Product of Array Except Self
 2. Maximum Product Pair
 3. Trapping Rain Water
 4. Maximum Sum Subarray
 5. Minimum Sum Subarray
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5. Collection & LINQ Programs (15 Questions)

1. Remove Duplicates Using HashSet
 2. Dictionary Frequency Count
 3. Sort List<T>
 4. Find Max Value Using LINQ
 5. Find Min Value Using LINQ
 6. Find Duplicates Using LINQ
 7. Distinct Values Using LINQ
 8. Find Second Highest Salary
 9. Employee Grouping by Department
 10. GroupBy Examples
 11. Top N Records
 12. Convert List to Dictionary
 13. Flatten Nested Collections
 14. Find Common Records
 15. Pagination and Dynamic Filtering
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6. Recursion Programs (15 Questions)

1. Factorial Using Recursion
2. Fibonacci Using Recursion
3. Sum of Digits Using Recursion
4. Reverse String Using Recursion
5. GCD Using Recursion
6. Binary Search Using Recursion
7. Power Calculation Using Recursion
8. Palindrome Using Recursion
9. Print Numbers Recursively
10. Generate Subsets
11. Generate Permutations
12. Tower of Hanoi
13. Rat in a Maze
14. N-Queen Problem
15. Sudoku Solver

7. Matrix Programs (15 Questions)

1. Matrix Addition
2. Matrix Subtraction
3. Matrix Multiplication
4. Matrix Transpose
5. Diagonal Sum
6. Boundary Elements
7. Spiral Matrix Traversal
8. Rotate Matrix 90 Degrees
9. Check Symmetric Matrix
10. Identity Matrix
11. Upper Triangular Matrix
12. Lower Triangular Matrix
13. Search in Matrix
14. Row Sum and Column Sum
15. Matrix Determinant

8. Linked List Programs (10 Questions)

1. Insert Node
2. Delete Node
3. Reverse Linked List
4. Find Middle Node
5. Find Nth Node from End
6. Detect Loop
7. Remove Loop
8. Merge Two Linked Lists
9. Remove Duplicates
10. Check Palindrome Linked List

9. Stack & Queue Programs (10 Questions)

1. Implement Stack
2. Implement Queue
3. Reverse Stack
4. Balanced Parentheses
5. Next Greater Element
6. Queue Using Two Stacks
7. Stack Using Two Queues
8. Circular Queue

9. Min Stack
 10. Evaluate Postfix Expression
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10. Interview Favorite Logical Programs (20 Questions)

Frequently Asked in Product-Based Companies

1. FizzBuzz
 2. Two Sum
 3. Longest Substring Without Repeating Characters
 4. Valid Parentheses
 5. Merge Intervals
 6. LRU Cache
 7. Find Duplicate Number
 8. Majority Element
 9. Longest Common Prefix

 10. Roman to Integer

 11. Integer to Roman

 12. Merge Sorted Arrays
 13. Remove Duplicates from Sorted Array
 14. Rotate Image
 15. Climbing Stairs
 16. House Robber
 17. Coin Change
 18. Word Break
 19. Longest Increasing Subsequence
 20. Maximum Depth of Binary Tree
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Most Important Questions for Senior .NET Developers

Focus on these first:

- Reverse String
- Palindrome
- Prime Number
- Fibonacci

- Armstrong Number
 - Anagram
 - Missing Number in Array
 - Duplicate Elements in Array
 - Two Sum
 - Kadane's Algorithm
 - Valid Parentheses
 - Second Highest Salary using LINQ
 - Employee Grouping using LINQ
 - Remove Duplicates using LINQ
 - Binary Search
 - Reverse Linked List
 - Detect Loop in Linked List
 - LRU Cache
 - Merge Intervals
 - Longest Substring Without Repeating Characters
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Recommended Study Plan

Week 1

- Number Programs
- Pattern Programs

Week 2

- String Programs

Week 3

- Array Programs

Week 4

- Collections and LINQ

Week 5

- Recursion and Matrix

Week 6

- Linked List, Stack, Queue

Week 7

- LeetCode Medium Problems

Week 8

- Mock Interviews
- Timed Coding Practice